TIMEWARP

A granular delay, looper & sampler built for sonic exploration.

Capture live audio or import samples, then independently stretch time and transpose pitch to create everything from subtle enhancements to lush soundscapes. Connect a MIDI device to achieve polyphony, transforming any sound source into a playable, multi-layered instrument.

Controls

Switches

1. **On / Off**

   Controls if the effect is bypassed.

2. **Record / Dub**

   Enable to record into the buffer.

3. **Play / Stop**

   Enable to hear sound played back from the buffer.

4. **Erase**

   When triggered, this will clear the buffer contents.

Granular controls

5. **Scan**

   Controls the start position for each grain.

6. **Spray**

   Controls the amount of random on the start position for each grain. Put this on zero if you want some glitchy, digital sounds. For a smooth sound put it on non-zero.

7. **Freeze**

   Sets the *Stretch* control to zero which effectively freezes the sample playback position.

8. **Stretch**

   Controls the start position of the grains. At zero the sample playback position is frozen. At -1 it will play back in reverse. At 2 it will play back at double the speed.

9. **Size**

   Controls the duration of the grains. At full clockwise the whole buffer will be played. The middle position gives you a grain size of 500 milliseconds. And at full counter-clockwise you will get the smallest possible grain size of 10 milliseconds.

10. **Density**

    Controls the amount of overlap between the grains. E.g. how many grains will be played simultaneously. At full counter-clockwise you will get one grain at a time. At full clockwise you will get 8 grains at a time.

11. **Stereo**

    Controls the stereo width of the grains. At full counter-clockwise each grain is mono. At full clockwise each grain is randomly hard-panned to either the left or right. Dial it in for a stereo width in-between these two.

12. **Detune**

    Controls the pitch of the grains in cents.

13. **Pitch**

    Controls the pitch of the grains in semitones.

Buffer controls

14. Sample

Determines which audio file is loaded from disk. It's only active if *Sample Mode* is set to the *Sampler* option.

15. Sample Mode

Controls how we write to the buffer. In *Delay* mode the delay time is set with the *Time* parameter. In *Looper* mode the delay time is set to the length of the first recording. In *Sampler* mode you can load a file from disk. The time is set to the length of the loaded file. You can record and overdub in each of these modes.

16. Time / Length

- *Time* is activated and visible if the *Sampler Mode* is set to *Delay*. *Time* sets the delay time in milliseconds.
- *Length* is activated and visible if the *Sampler Mode* is set to *Looper* or *Sampler*. Like the *Time* parameter this controls the delay time. But in this case it's a fractional value of the sample or loop duration.

17. Highpass & Lowpass

Highpass and *Lowpass* set the filter cutoff frequencies in hertz. The filter is in the feedback path. It's only activated when you're recording or overdubbing.

18. Recycle

Blends the original and processed signals within the feedback loop. At full counter-clockwise the signal is preserved as-is. At full clockwise the wet signal including all the pitch shifting and time stretching effects are fed back. The *Feedback* parameter controls how much of the signal is actually being fed back. So if *Feedback* is set to zero, *Recycle* will have no audible effect."

19. Feedback

Controls how much of the signal is fed back when overdubbing.

MIDI controls

20. Attack, Decay, Sustain & Release

Controls the amplitude envelope for each voice.

21. MIDI

Enables or disables MIDI input. When enabled, MIDI notes and pitchbend messages control the pitch. Velocity controls the volume. And sustain will hold the notes until released.

22. Sync Position

Controls whether multiple notes will be synchronized to the same playback position or not. If it's on the first active note will play back from the beginning of the buffer. Notes that are triggered later will follow the position of that first note. If it's off, each note will always start playing back from the beginning of the buffer.

23. Voices

Controls how many notes can sound simultaneously. The maximum is 8 voices.

Mix controls

24. Dry

Controls the dry signal volume.

25. Wet

Controls the effect signal volume.